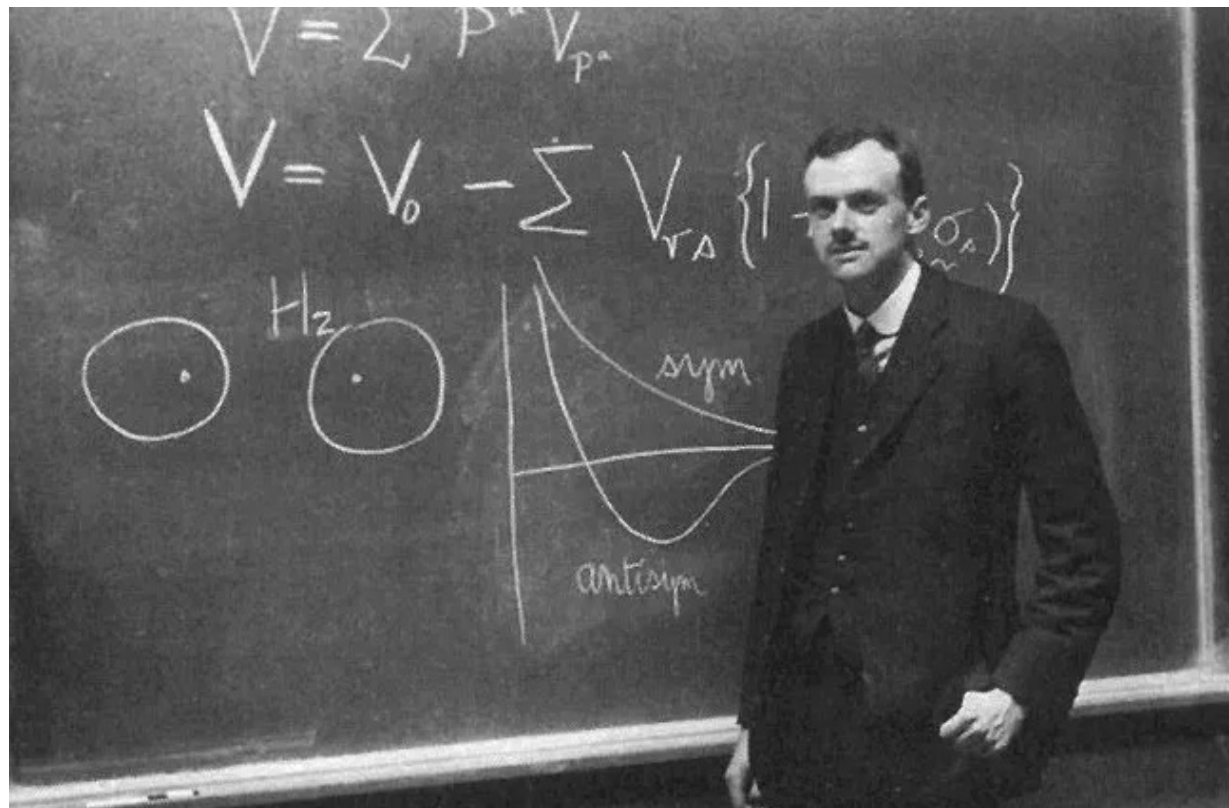


ALGEBRA DE DIRAC



Notación de DIRAC

$|\phi\rangle \rightarrow$ ket Estado vector

$\langle\phi| = |\phi\rangle^\dagger$: Bra

$\langle x|\phi\rangle$ Bracket

$$\langle x|\phi\rangle \equiv (\langle x|, |\phi\rangle)$$

$\{|i\rangle\}$ BASE

$$\hat{I} = \sum_i |i\rangle\langle i| \quad \text{operador unidad}$$

Caracteriza la propiedad de completitud de una base.

$$|\phi\rangle \equiv \hat{I}|\phi\rangle = \sum_i |i\rangle\langle i|\phi\rangle = \sum_i c_i |i\rangle$$

$$\langle\phi| = \sum_i c_i^* \langle i|$$

$$\langle\phi|\hat{I}|\phi\rangle = \sum_i \underbrace{\langle\phi|i\rangle}_{c_i^*} \underbrace{\langle i|\phi\rangle}_{c_i}$$

$$\begin{aligned} \langle\phi|\phi\rangle &= \sum_{ij} c_i c_j^* \underbrace{\langle j|i\rangle}_{\delta_{ij}} \\ &= \sum_j |c_j|^2 = \|\phi\|^2 \end{aligned}$$

$$\begin{aligned} |\psi\rangle &= \hat{I}|\psi\rangle = \sum_i |i\rangle\langle i|\psi\rangle \\ &= \sum_i D_i |i\rangle \end{aligned}$$

$$\text{AMPLITUD } \langle x|\hat{A}|\phi\rangle$$

$$\begin{aligned} \langle x|\hat{I}\hat{A}\hat{I}|\phi\rangle &= \sum_i \sum_j \underbrace{\langle x|i\rangle}_{E_i^*} \underbrace{\langle i|\hat{A}|j\rangle}_{A_{ij}} \underbrace{\langle j|\phi\rangle}_{c_j} \\ &= \sum_i \sum_j E_i^* c_j A_{ij} \end{aligned}$$

$$BA = C$$

$$\begin{aligned} \langle x|BA|\phi\rangle &= \langle x|\hat{I}\hat{B}\hat{I}\hat{A}\hat{I}|\phi\rangle \\ &= \sum_{ijk} \underbrace{\langle x|i\rangle}_{E_i^*} \underbrace{\langle i|\hat{B}|j\rangle}_{B_{ij}} \underbrace{\langle j|\hat{A}|k\rangle}_{A_{jk}} \underbrace{\langle k|\phi\rangle}_{c_k} \end{aligned}$$

Ejm. Expresc A en términos de sus elementos matriciales.

$$\begin{aligned} A &= \hat{1} A \hat{1} \\ &= \sum_{ij} |i\rangle \underbrace{\langle i|A|j\rangle}_{A_{ij}} \langle j| \end{aligned}$$

$$A = \sum_{ij} A_{ij} |i\rangle \langle j|$$

↑
elementos matriciales

Representación explícita de estados y operadores

ket $\rightarrow$ vector columna

Estado base $\{|i\rangle\} = \{|1\rangle, |2\rangle\}$

$$\langle i|j\rangle = \delta_{ij}$$

$$|1\rangle = \hat{1}|1\rangle = \sum_{i=1}^2 |i\rangle \langle i|1\rangle$$

$$= |1\rangle \langle 1|1\rangle + |2\rangle \langle 2|1\rangle$$

$$= 1|1\rangle + 0|2\rangle$$

$$|1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |2\rangle \rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$|\phi\rangle = \hat{1}|\phi\rangle = \sum |i\rangle \langle i|\phi\rangle$$

$$= |1\rangle \langle 1|\phi\rangle + |2\rangle \langle 2|\phi\rangle$$

$$= \phi_1 |1\rangle + \phi_2 |2\rangle$$

$$|\phi\rangle \rightarrow \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

$$\langle \phi | = \phi_1^* \langle 1 | + \phi_2^* \langle 2 |$$

$$= (\phi_1^*, \phi_2^*)$$

vector fila (Bra)

$$\langle X | \phi \rangle = (X_1^* \quad X_2^*) \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

$$= X_1^* \phi_1 + X_2^* \phi_2$$

$$\hat{1} = \sum_i |i\rangle \langle i| = |1\rangle \langle 1| + |2\rangle \langle 2|$$

$$= \begin{pmatrix} 1 \\ 0 \end{pmatrix} (1 \ 0) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} (0 \ 1)$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\hat{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Ejm. Expresa en forma matricial $|\psi\rangle = A|\phi\rangle$

$$\langle j | \psi \rangle = \langle j | A \hat{1} | \phi \rangle$$

$$= \sum_i \langle j | A | i \rangle \langle i | \phi \rangle$$

$$\psi_j = \sum_i A_{ji} \phi_i$$

$$\psi_1 = A_{11} \phi_1 + A_{12} \phi_2$$

$$\psi_2 = A_{21} \phi_1 + A_{22} \phi_2$$

$$\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

Base continua

$$(\varphi, \psi) = \int d^3 \bar{r} \varphi^*(\bar{r}) \psi(\bar{r})$$

$$(|\varphi\rangle, |\psi\rangle) = \langle \varphi | \psi \rangle$$

BRACKET

$$\{|\bar{r}\rangle\} \quad \{|\bar{p}\rangle\}$$

Dos representaciones

$$\xi_{\bar{r}_0}(\bar{r}) = \delta(\bar{r} - \bar{r}_0)$$

$$\xi_{\bar{r}_0}(\bar{r}) \rightarrow |\bar{r}_0\rangle \text{ (deltas)}$$

$$\psi_{\bar{p}_0}(\bar{r}) \rightarrow |\bar{p}_0\rangle \text{ (ondas planas)}$$

Operadores Unidad:

$$\int d^3 \bar{r}_0 |\bar{r}_0\rangle \langle \bar{r}_0| = \mathbb{1}$$

$$\int d^3 \bar{p}_0 |\bar{p}_0\rangle \langle \bar{p}_0| = \mathbb{1}$$

$$\langle \bar{r}_0 | \bar{p}_0 \rangle = (2\pi\hbar)^{-3/2} e^{\frac{i}{\hbar} \bar{p}_0 \cdot \bar{r}_0}$$

Componentes de un ket

$$|\psi\rangle = \int d^3 \bar{r}_0 |\bar{r}_0\rangle \underbrace{\langle \bar{r}_0 | \psi \rangle}_{\psi(\bar{r}_0)}$$

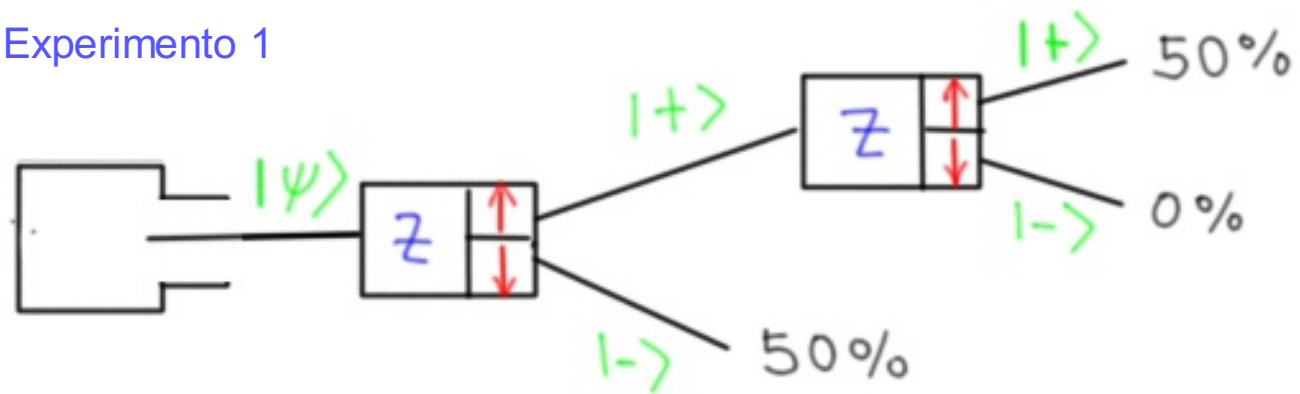
$$|\psi\rangle = \int d^3 \bar{p}_0 |\bar{p}_0\rangle \underbrace{\langle \bar{p}_0 | \psi \rangle}_{\tilde{\psi}(\bar{p}_0)}$$

$|\psi\rangle$ se puede expresar
en la base $\{|\bar{r}_0\rangle\}$ o
en la base $\{|\bar{p}_0\rangle\}$

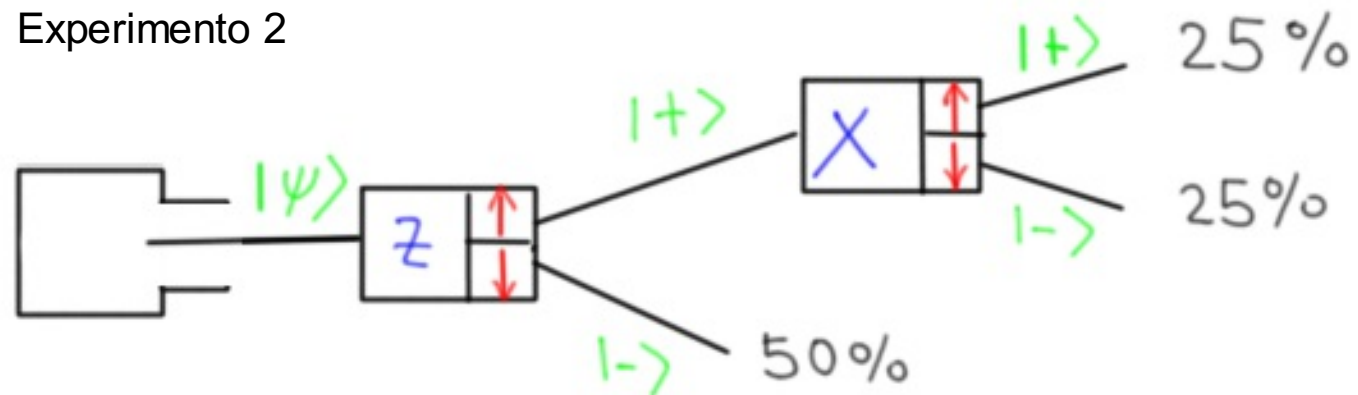
AYANT

Experimentos de Stern Gerlach

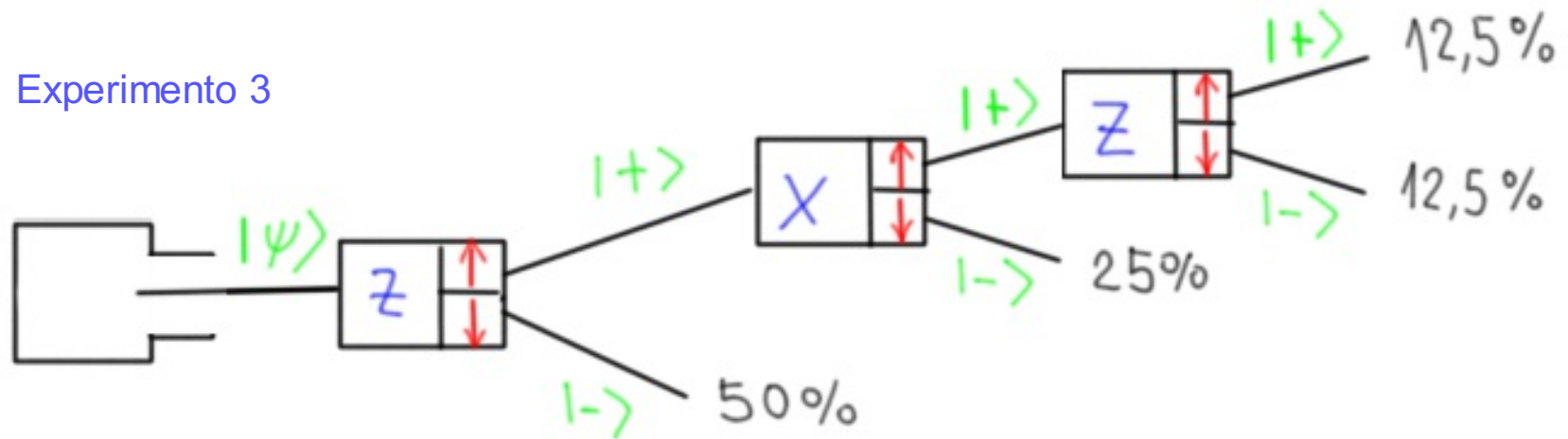
Experimento 1



Experimento 2



Experimento 3



Experimento 4

